Cell Line Ontology in 2026

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Abstract

Background: Cell lines are widely used in biomedical research, including studies of vaccine mechanisms, cellular immune responses following vaccination, and the development of vaccines and vaccine adjuvants. As a member of the OBO Foundry library, the Cell Line Ontology (CLO) was initiated in 2010 to standardize the representation of cell lines and integrate cell line information from multiple sources, including ATCC, HyperCLDB, and other cell repositories. As of June 24, 2024, CLO contained more than 43,000 terms, including 39,115 CLO-specific terms and over 38,000 cell line cell terms, including many immune cell lines used in vaccine development and cancer immunotherapy studies. CLO has been widely used for biomedical data annotation and cell line–related data analysis; however, its active development has been limited since 2024.

Methods: In 2026, we resumed CLO development and identified several major issues. First, ROBOT, an ontology development and quality-checking tool, reported 609 errors, including 83 duplicate labels, 507 duplicate definitions, and 6 terms with multiple definitions. Second, some imported terms from source ontologies had been deprecated or replaced. For example, 56 imported terms from the Human Disease Ontology (DOID) no longer existed in DOID, 2 imported Cell Ontology (CL) terms had been replaced by Dictyostelium discoideum Anatomy Ontology (DDANAT) terms, and 8 imported CL terms had been deprecated. Third, the design pattern for representing cell line cells was not implemented consistently across CLO. These issues were addressed primarily through manual editing of CLO. OntoFox tool was used to import terms from external source ontologies, and ROBOT was used for error checking and conversion of tab-delimited template files into OWL format.

Results: To address duplicate labels, we identified four pairs of cell line cell terms with identical labels that represented distinct cell lines and revised their labels accordingly. The remaining duplicate-label terms represented the same cell line and were therefore merged, with one term retained and redundant terms deprecated. In total, 38 CLO terms were deprecated during this process. We also revised definitions to resolve duplicate-definition and multiple-definition errors and fixed additional issues related to deprecated-term labels, ontology licensing, and other quality-control checks. As a result, all 609 ROBOT-reported errors were resolved.

We also updated imported terms and the associated CLO terms that referenced them, ensuring that deprecated external terms are no longer reused in CLO. In addition, we improved the modeling of existing cell line cell terms by explicitly representing, where applicable, whether a cell line cell is derived from another cell line cell, a specific cell type, a tissue or organ part, or a specific organism, using appropriate ontology relations. ROBOT templates were used to generate cell line cell terms, supporting consistent logical representation in OWL format.

CLO was originally built on the Basic Formal Ontology (BFO). Recently, the OBO Foundry community developed the Core Ontology for Biology and Biomedicine (COB), which is based on BFO and incorporates key terms from multiple OBO Library ontologies to improve interoperability and reuse. We therefore transformed CLO to align with COB, strengthening its interoperability with other OBO ontologies. We also implemented a Makefile-based workflow for importing terms from external source ontologies using OntoFox, generating OWL files from tab-delimited files using ROBOT templates, and building ontology release files.

Conclusion: These updates enable more efficient and consistent CLO development and support queries over cell lines based on multiple features, including derived cell type, derived anatomical structure, source organism, disease model, and cell line repository. Future work will focus on incorporating additional cell lines from resources such as ATCC and Cellosaurus to better support cell line–based biomedical data annotation and analysis.

Keywords

Ontology, Cell Line Ontology, cell line

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